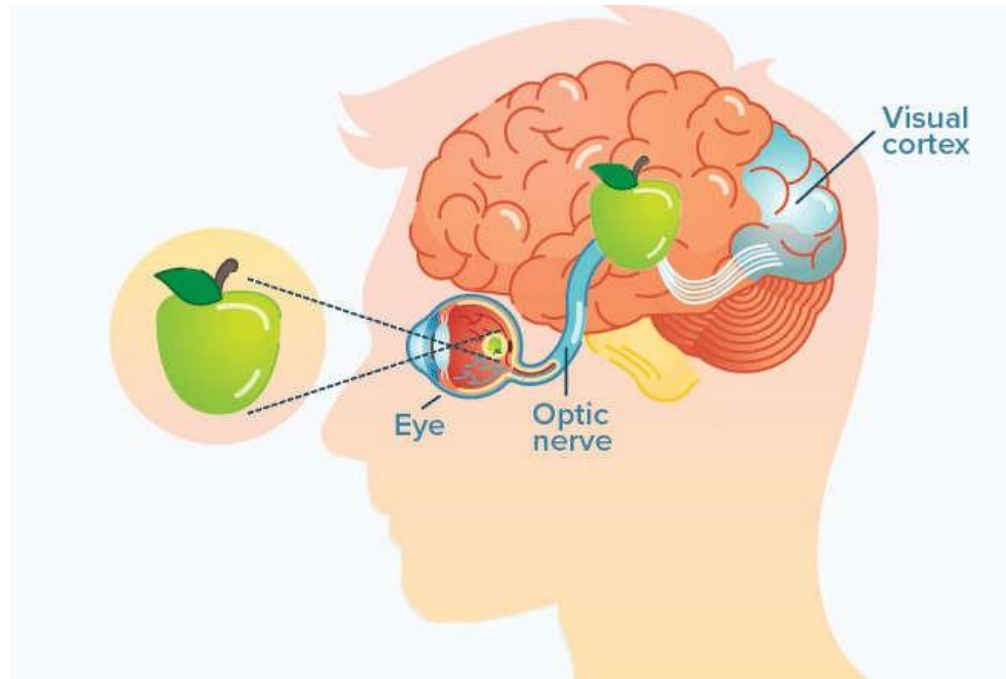


Introduction to Data Science



Preattentive Attributes

We see with our brains



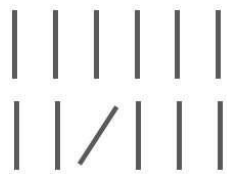
Types of memory

- **Iconic memory** is super fast, happens unconsciously, and is stimulated when we look at the world around us.
 - It is tuned to a set of preattentive attributes.
- **Short-term memory** has limitations. Specifically, people can keep about **four** chunks of visual information in their short-term memory at a given time.
- **Long-term memory** is built up over a lifetime and is vitally important for pattern recognition and general cognitive processing.

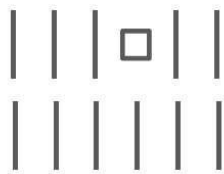
Iconic Memory

756**3**9506847**3**
65866**3**0**3**7576
860**3**72658602
8465891078**3**0

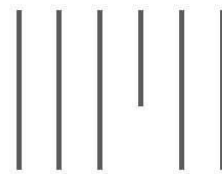
Preattentive Attributes



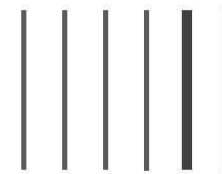
Orientation



Shape



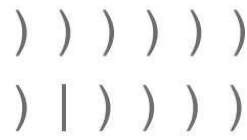
Line length



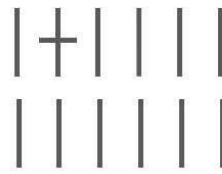
Line width



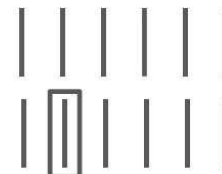
Size



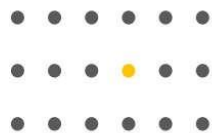
Curvature



Added marks



Enclosure



Hue



Intensity



Spatial position



Motion

Preattentive Attributes

- If we use preattentive attributes strategically, they can help us *enable our audience to see what we want them to see before they even know they're seeing it!*
- Preattentive attributes can:
 - direct your audience's attention to where you want
 - be used to create a visual hierarchy of elements

Preattentive Attributes in Text

No preattentive attributes

What are we doing well? Great Products. These products are clearly the best in their class. Replacement parts are shipped when needed. You sent me gaskets without me having to ask. Problems are resolved promptly. Bev in the billing office was quick to resolve a billing issue I had. General customer service exceeds expectations. The account manager even called to check in after normal business hours.
You have a great company – keep up the good work!

Color

What are we doing well? Great Products. **These products are clearly the best in their class.** Replacement parts are shipped when needed. You sent me gaskets without me having to ask. Problems are resolved promptly. Bev in the billing office was quick to resolve a billing issue I had. General customer service exceeds expectations. The account manager even called to check in after normal business hours.
You have a great company – keep up the good work!

Bold

What are we doing well? Great Products. These products are clearly the best in their class. Replacement parts are shipped when needed. You sent me gaskets without me having to ask. Problems are resolved promptly. Bev in the billing office was quick to resolve a billing issue I had. General customer service exceeds expectations. The account manager even called to check in after normal business hours.
You have a great company – keep up the good work!

Italics

What are we doing well? Great Products. These products are clearly the best in their class. *Replacement parts are shipped when needed.* You sent me gaskets without me having to ask. Problems are resolved promptly. Bev in the billing office was quick to resolve a billing issue I had. General customer service exceeds expectations. The account manager even called to check in after normal business hours.
You have a great company – keep up the good work!

Preattentive Attributes in Text

Size

What are we doing well? Great Products. These products are the best in their class. Replacement parts are shipped when needed. You sent gaskets

without me having to

ask. Problems are resolved promptly. Bev in the billing office was quick to resolve a billing issue I had. General customer service exceeds expectations. The account manager even called to check in after normal business hours. You have a great company – keep up the good work!

Separate spatially

What are we doing well? Great Products. These products are clearly the best in their class. Replacement parts are shipped when needed. You sent me gaskets without me having to ask.

Problems are resolved promptly.

Bev in the billing office was quick to resolve a billing issue I had. General customer service exceeds expectations. The account manager even called to check in after normal business hours. You have a great company – keep up the good work!

Outline (enclosure)

What are we doing well? Great Products. These products are clearly the best in their class. Replacement parts are shipped when needed. You sent me gaskets without me having to ask. Problems are resolved promptly. Bev in the billing office was quick to resolve a billing issue I had. General customer service exceeds expectations. The account manager even called to check in after normal business hours.

You have a great company – keep up the good work!

Underline (added marks)

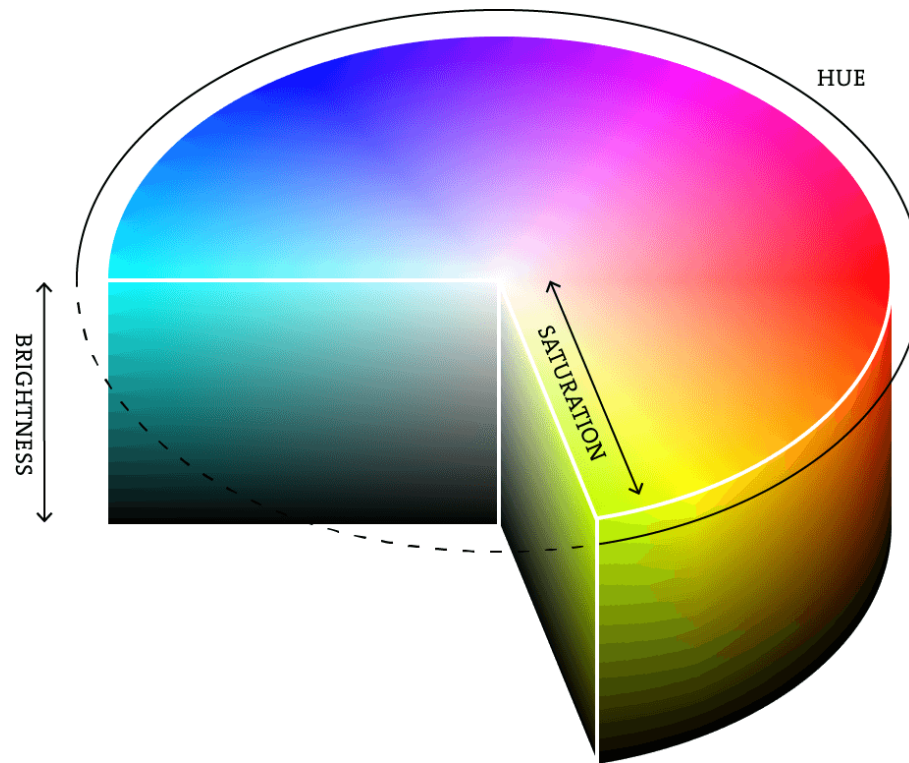
What are we doing well? Great Products. These products are clearly the best in their class. Replacement parts are shipped when needed. You sent me gaskets without me having to ask. Problems are resolved promptly. Bev in the billing office was quick to resolve a billing issue I had. General customer service exceeds expectations. The account manager even called to check in after normal business hours.

You have a great company – keep up the good work!

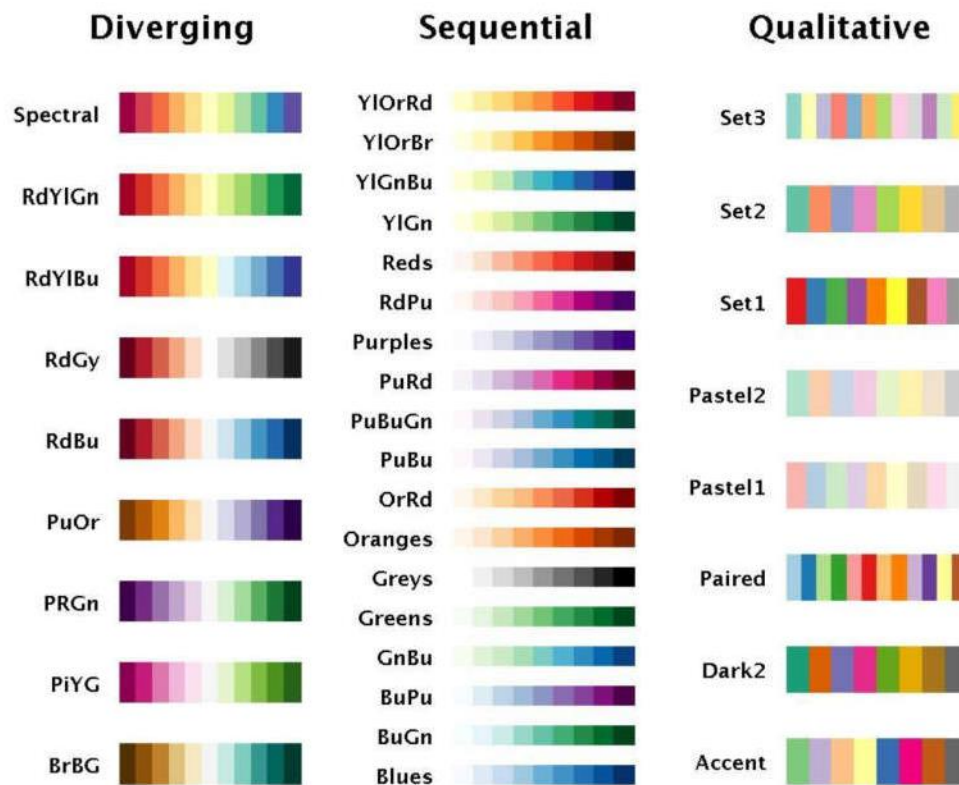
Color

- Color is one of the most powerful tools you have for drawing your audience's attention.
- Resist the urge to use color for the sake of being colorful.
- Leverage color selectively as a strategic tool to highlight the important parts of your visual.
- The use of color should always be an intentional decision.
 - Never let your tool make this important decision for you!

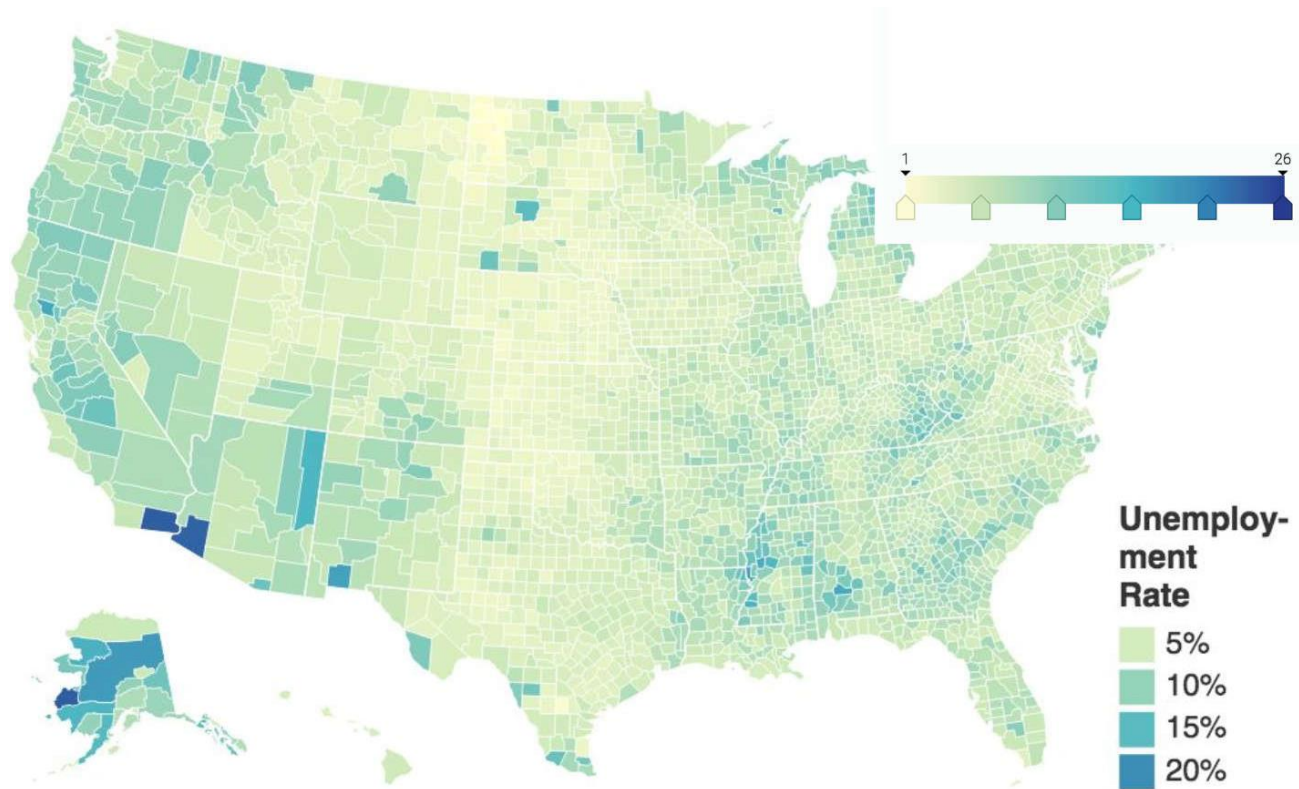
HSB Model



Color Scales (colorbrewer.org)



Mapping Color Scales



Use Color Sparingly

Country Level Sales Rank Top 5 Drugs

Rainbow distribution in color indicates sales rank in given country from #1 (red) to #10 or higher (dark purple)

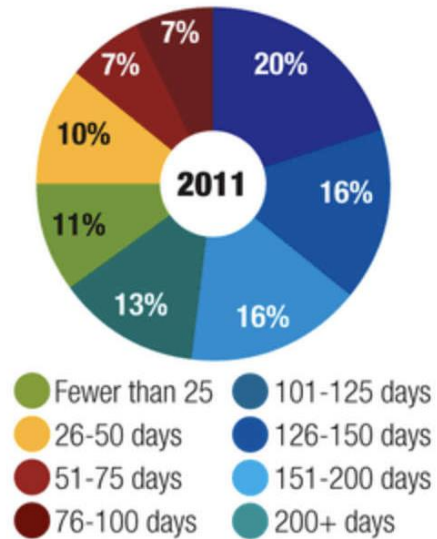
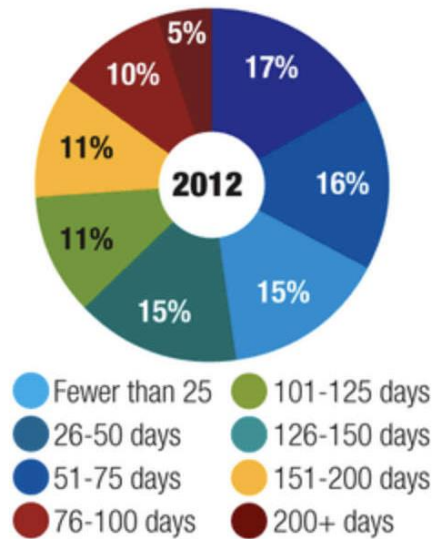
Country	A	B	C	D	E
AUS	1	2	3	6	7
BRA	1	3	4	5	6
CAN	2	3	6	12	8
CHI	1	2	8	4	7
FRA	3	2	4	8	10
GER	3	1	6	5	4
IND	4	1	8	10	5
ITA	2	4	10	9	8
MEX	1	5	4	6	3
RUS	4	3	7	9	12
SPA	2	3	4	5	11
TUR	7	2	3	4	8
UK	1	2	3	6	7
US	1	2	4	3	5

Top 5 drugs: country-level sales rank

RANK		1	2	3	4	5+
COUNTRY DRUG		A	B	C	D	E
Australia		1	2	3	6	7
Brazil		1	3	4	5	6
Canada		2	3	6	12	8
China		1	2	8	4	7
France		3	2	4	8	10
Germany		3	1	6	5	4
India		4	1	8	10	5
Italy		2	4	10	9	8
Mexico		1	5	4	6	3
Russia		4	3	7	9	12
Spain		2	3	4	5	11
Turkey		7	2	3	4	8
United Kingdom		1	2	3	6	7
United States		1	2	4	3	5

Use Color Consistently

How many nights do you typically stay away from home due to work?

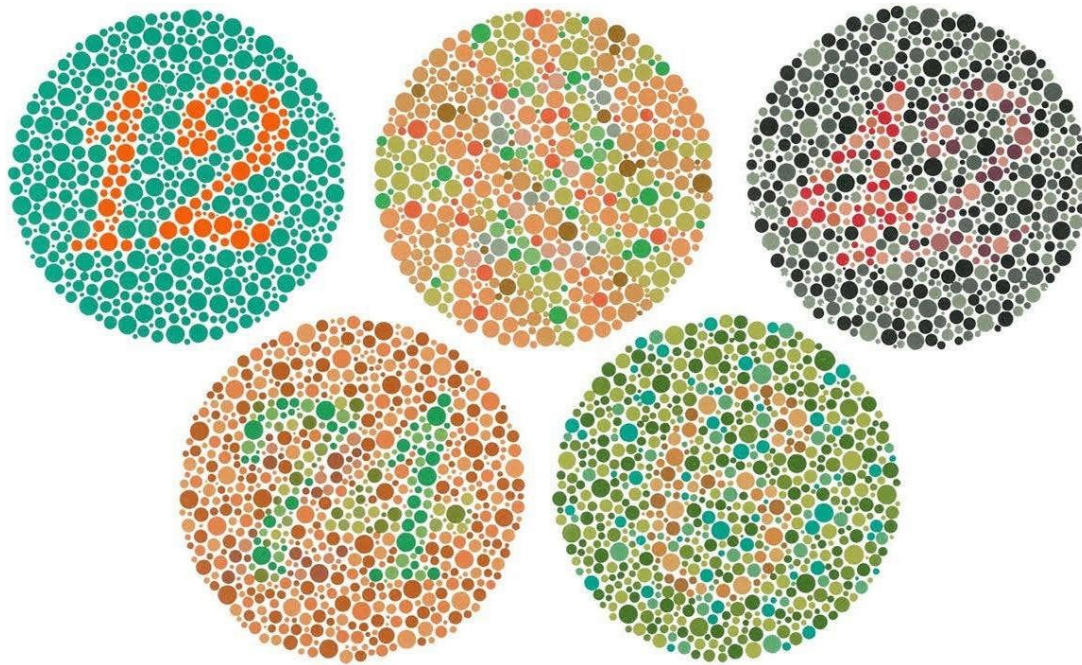


Who Took The Survey?

Overall, 170 consultants took our annual Best Places to Stay survey in the spring of 2012.



Design with colorblind in mind

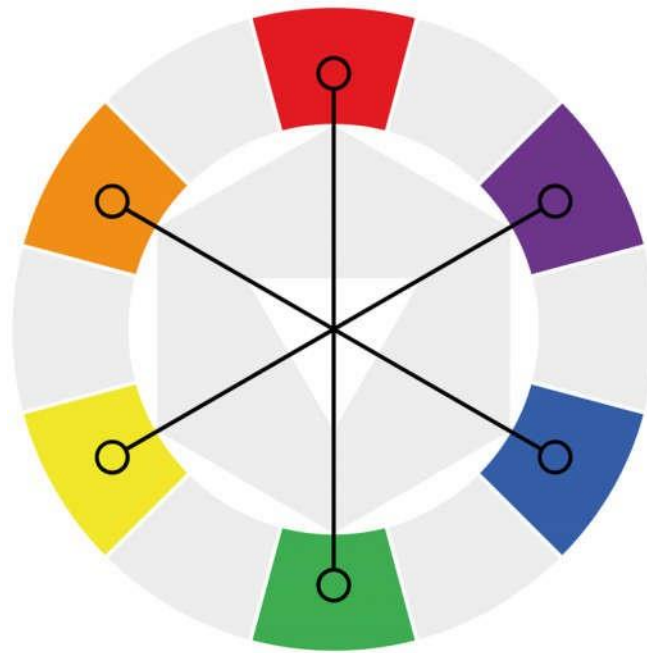


vischeck.com

Consider the tone color conveys

Red Power Danger Exciting Anger Strength Rebellious Love Violence Passion Warning Energy	Orange Energy Deceitful Enthusiastic Anger Creativity Overpowering Friendliness Warning Positivity	Yellow Optimistic Sickness Exciting Anxiety Energy Warning Happiness Cowardice Positivity	Green Health Materialistic Wealth Greed Hope Envy Growth Hazardous Fresh Sickness Nurturing
Blue Trust Sadness Loyalty Loneliness Peace Rigidity Calm Passiveness Stability Coldness	Purple Luxury Arrogance Spirituality Superiority Quality Moodiness Sophistication Introversiion Creative Suppression	Pink Romance Neediness Love Immature Nurturing Naivety Sweetness	Brown Earthiness Boring Comforting Predictable Wholesome Dull Approachable Lack of humor Stable
Black Sophistication Depressing Elegant Fear Power Negative Mystery Death Strength Evil Seduction Secretive	Grey Impartial Indecisive Professional Indifferent Mature Depressing Classic Negative Stable Isolation Elegant Lack of energy	White Innocence Plain Purity Emptiness Cleanliness Isolation Freshness Sterility Simplicity Distant	

Complementary Colors



Where are your eyes drawn?



Where are your eyes drawn?



Where are your eyes drawn?



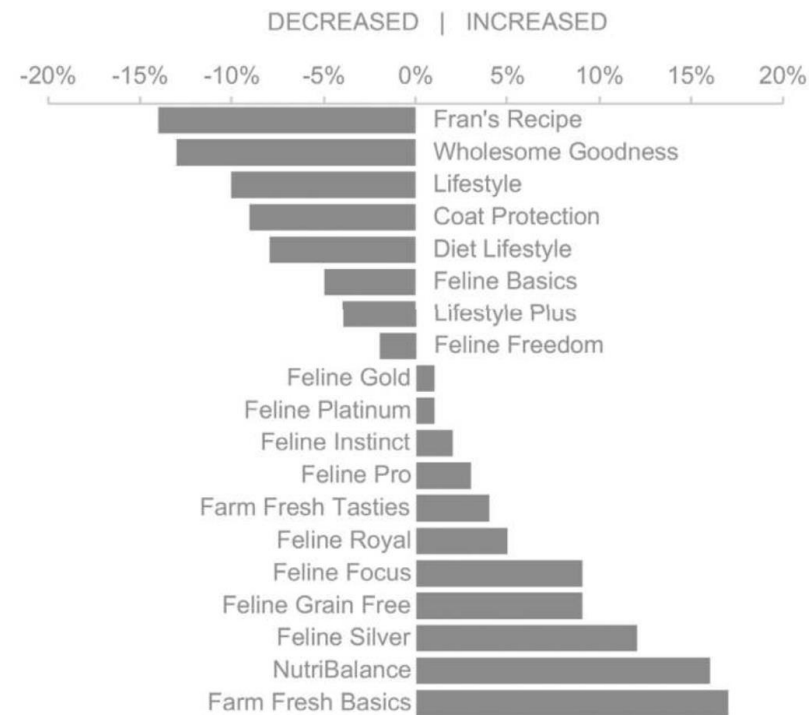
Where are your eyes drawn?



Focus Attention

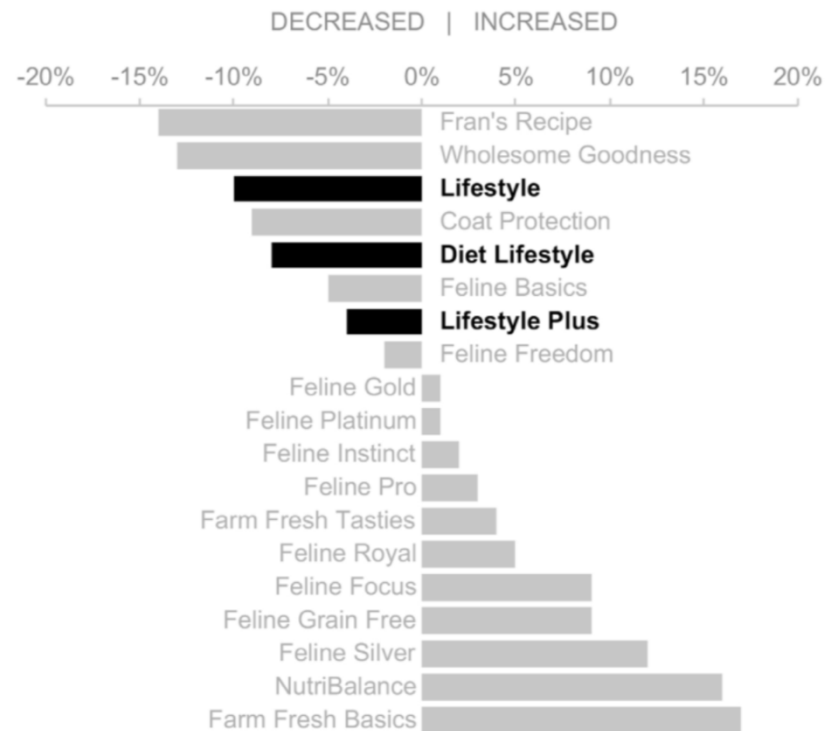
Cat food brands: YoY sales change

% CHANGE IN VOLUME (\$)



Focus Attention

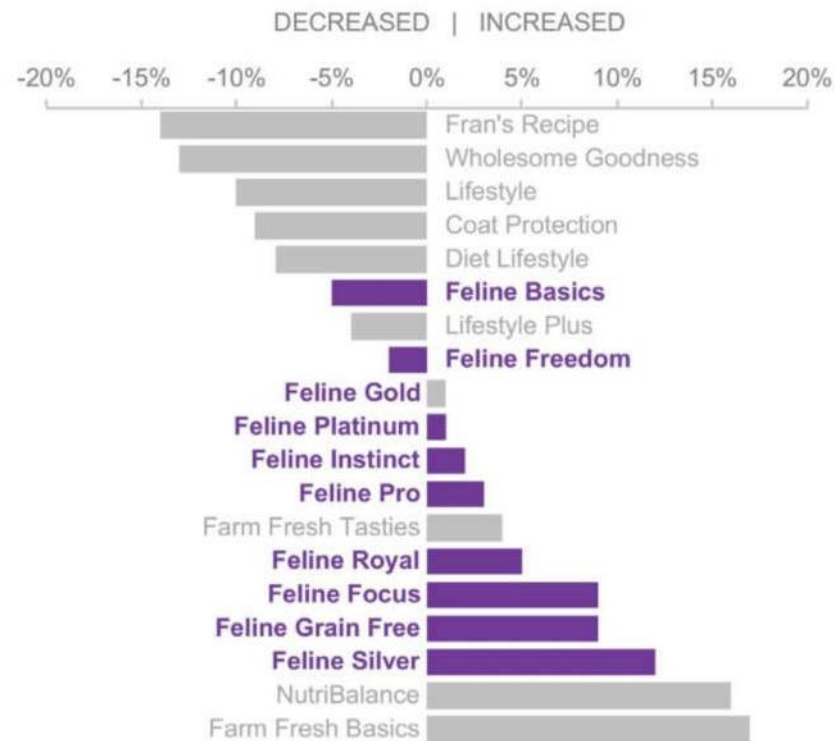
Cat food brands: **Lifestyle line brands declined**
YEAR-OVER-YEAR % CHANGE IN SALES VOLUME (\$)



Focus Attention

Cat food brands: **most in Feline line increased**

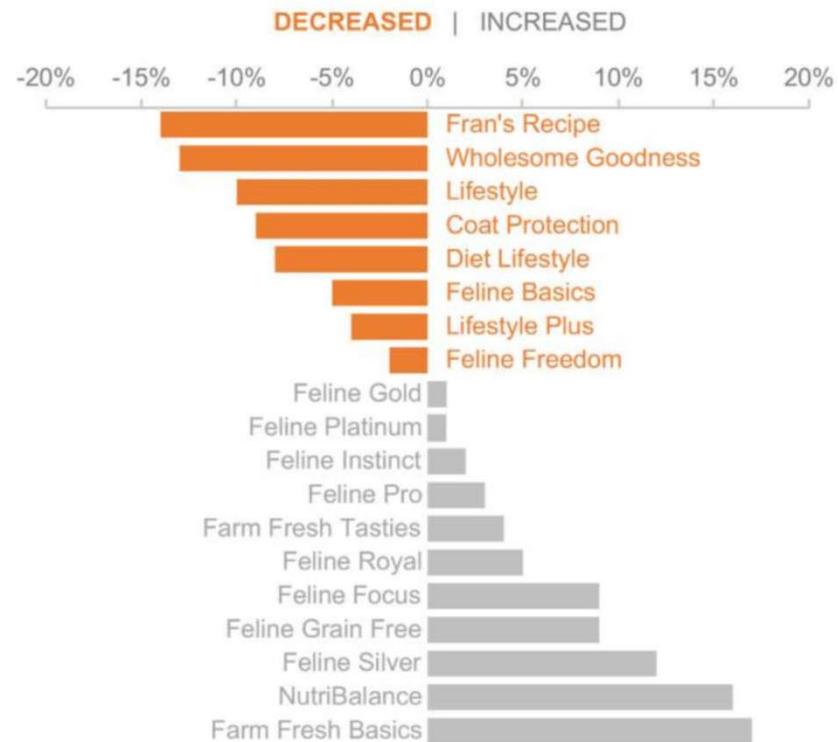
YEAR-OVER-YEAR % CHANGE IN SALES VOLUME (\$)



Focus Attention

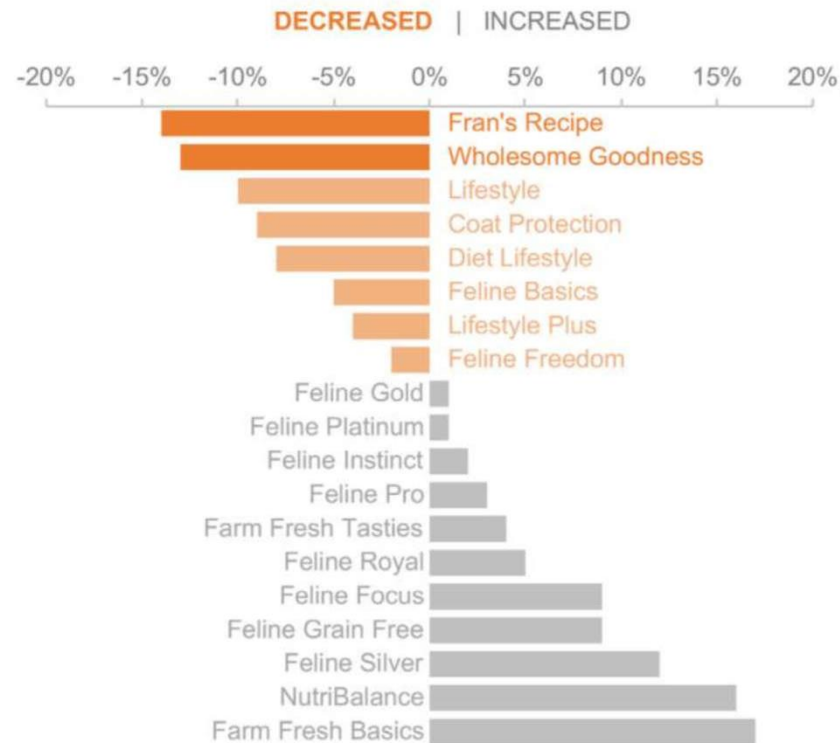
Cat food brands: **8 brands decreased in sales**

YEAR-OVER-YEAR % CHANGE IN SALES VOLUME (\$)



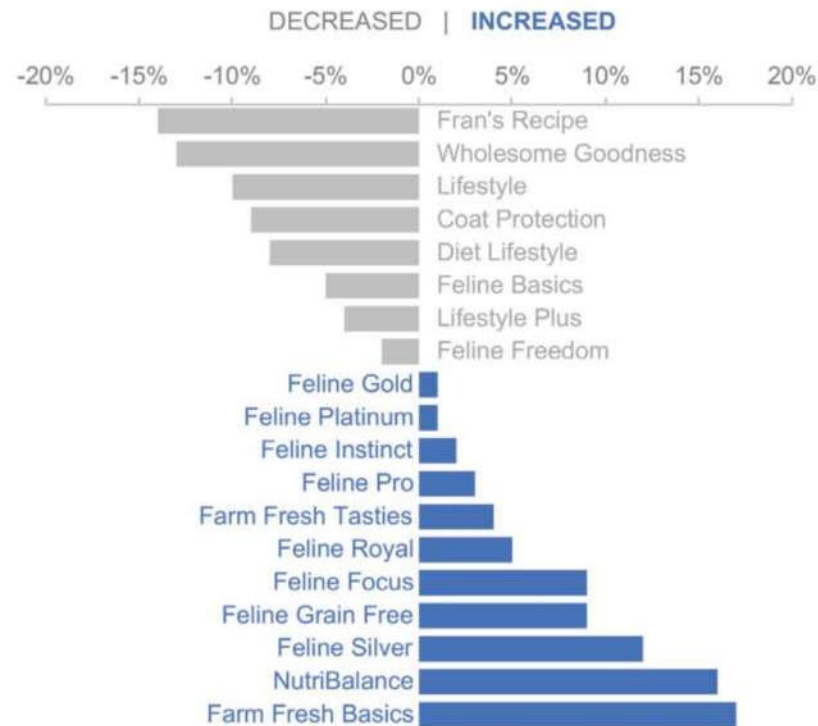
Focus Attention

Cat food brands: **2 brands decreased the most**
YEAR-OVER-YEAR % CHANGE IN SALES VOLUME (\$)



Focus Attention

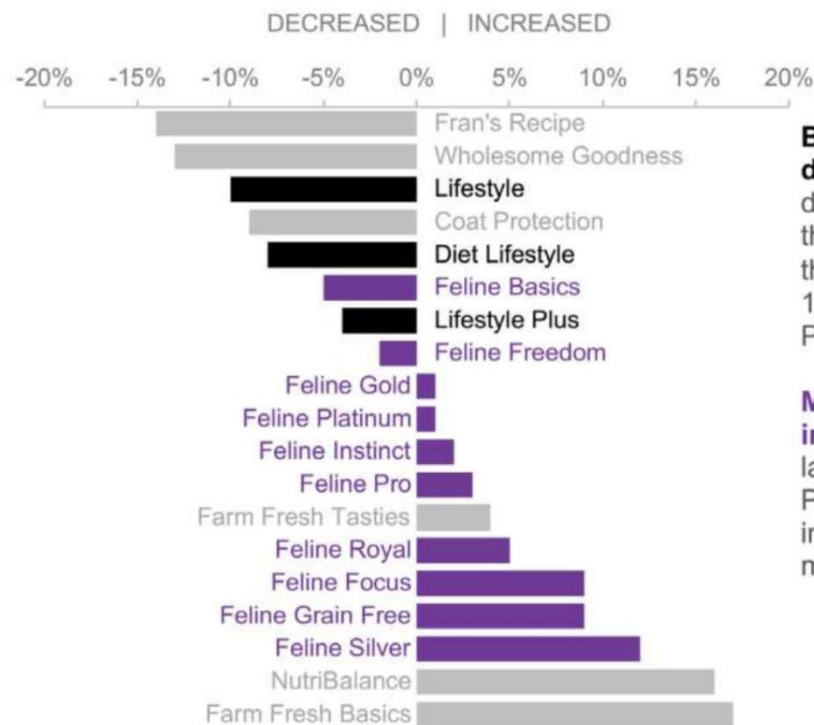
Cat food brands: **11 brands flat to increasing**
YEAR-OVER-YEAR % CHANGE IN SALES VOLUME (\$)



Focus Attention

Cat food brands: mixed results in sales year-over-year

YEAR-OVER-YEAR % CHANGE IN SALES VOLUME (\$)



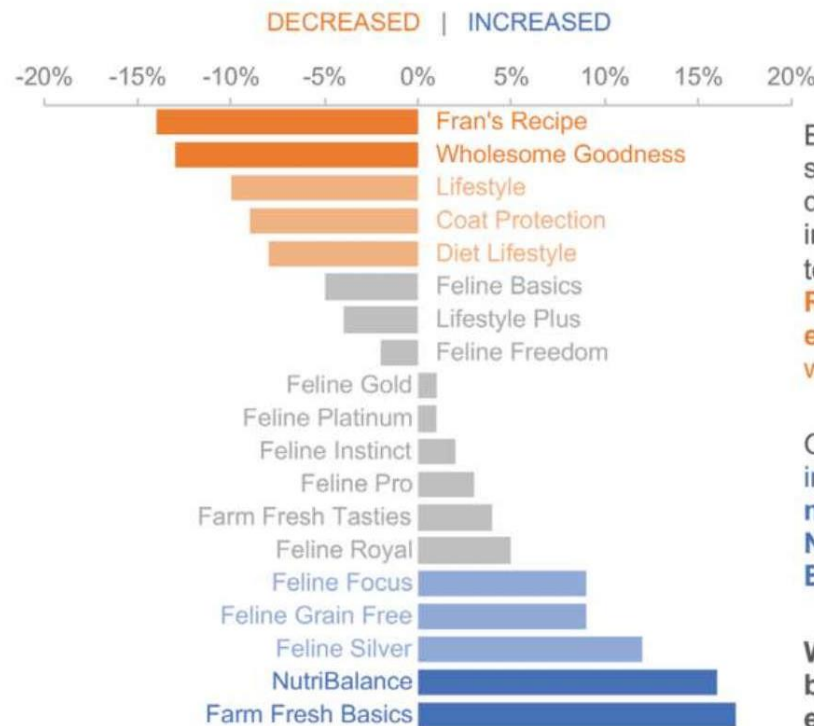
Brands in the Lifestyle line all decreased year-over-year, mainly due to a marketing shift away from these products. Classic Lifestyle had the biggest decrease in sales, down 10% year-over-year, while Lifestyle Plus had the smallest decrease (4%).

Most brands in the Feline line increased in sales year-over-year, largely due to the partnership with PetFriends retailers that we entered into mid-year. We anticipate continued momentum in the coming year.

Focus Attention

Cat food brands: mixed results in sales year-over-year

YEAR-OVER-YEAR % CHANGE IN SALES VOLUME (\$)

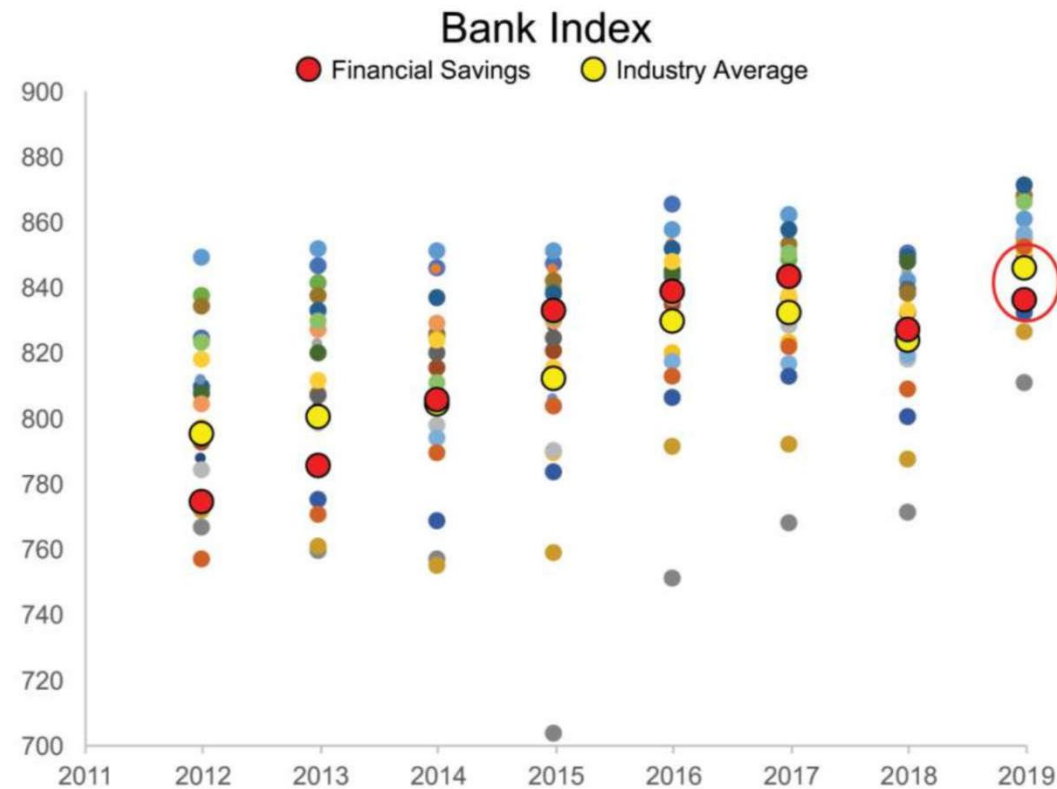


Eight key cat food brands declined in sales year-over-year, with five brands decreasing 7%+. This was expected in some cases due to focus shift toward higher margin brands. **Fran's Recipe and Wholesome Goodness each declined by more than 13%, which was more than expected.**

On the positive side, five brands increased 8%+ year-over-year, with **marked 16%+ increases for NutriBalance and Farm Fresh Basics.**

What can we learn from increasing brands that we can apply elsewhere? Let's discuss next steps.

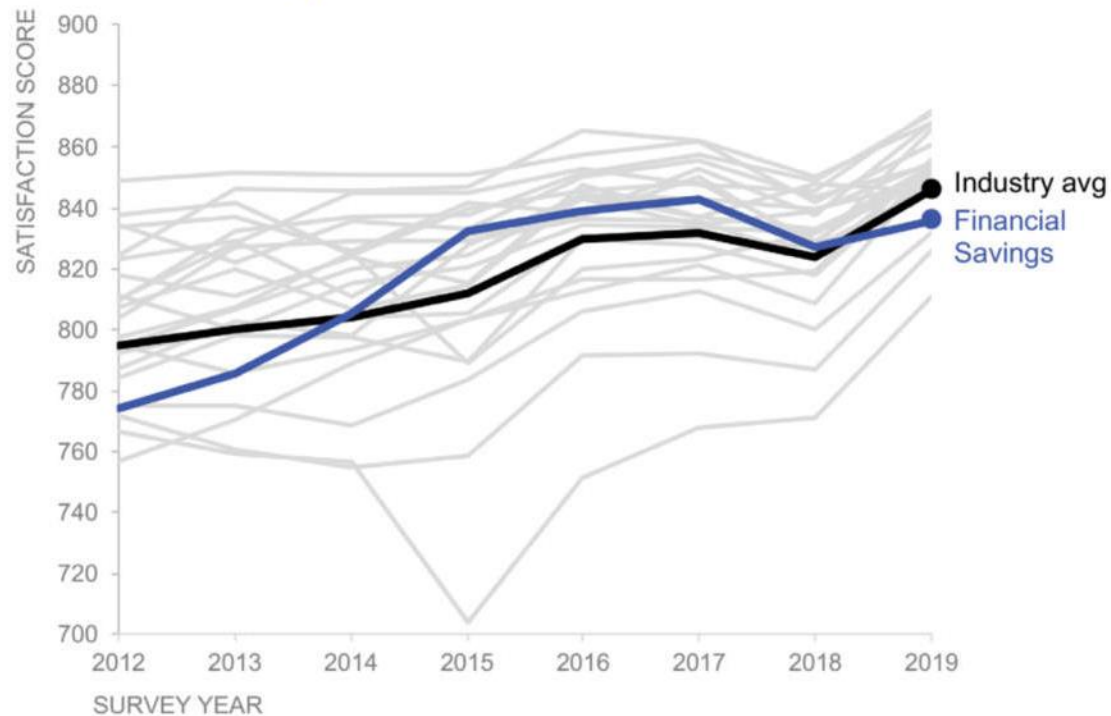
Visualize All the Data



Visualize All the Data

BRANCH SATISFACTION

Financial Savings below **industry** for first time in 5 years



Visualize All the Data

BRANCH SATISFACTION

Financial Savings below **industry** for first time in 5 years

